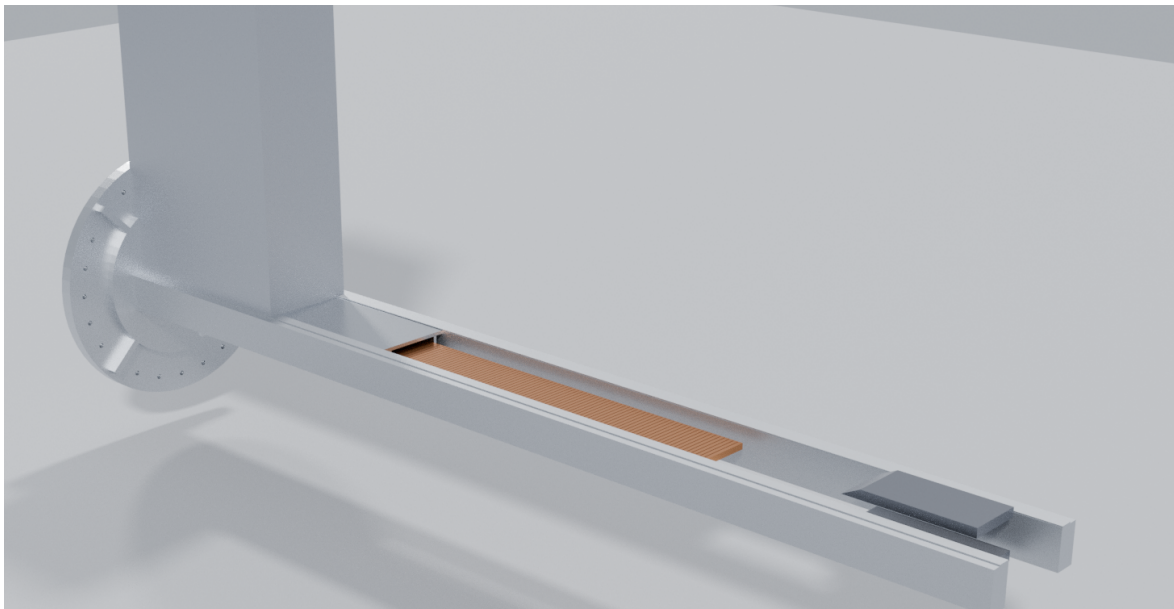


VOLLEY

An engineering study of **commanded, per-satellite orbit change** for CubeSats that are never modified.

Adityavardhan Mishra
B.Tech Mechanical Engineering
Symbiosis Institute of Technology
Symbiosis International (Deemed University), Pune
adityavardhanmishra@gmail.com

A CubeSat flown as a rideshare secondary inherits the primary customer's orbit. The spring that ejects it adds 1–2 m/s: enough for separation, and a real but very small change in orbital energy — far too little for commanded orbit shaping — and of the more than **4 800 nanosatellites and CubeSats catalogued** as of January 2026, on the order of **222 carry a propulsion system**, so the rest stay where they were dropped for life. VOLLEY asks what it costs to give each of them a *commanded* departure velocity without putting anything on the satellite.



Gen5 analysed architecture — CAD model, not flight hardware. Ironless double-sided Halbach linear synchronous motor, 1.5 m track, reusable magnet sled, twelve 3U satellites from two transverse cassettes, contactless eddy-current arrest. Nothing shown here has been built, fired or measured.

THE PROGRAMME, AND WHAT EACH LAYER IS WORTH

VOLLEY	the programme	one question, asked to a fixed evidence standard: can a propulsion-less satellite be given its own orbit?
Gen5	the analysed evidence base	a self-contained electromagnetic deployer. 64 run-sheet files covering 61 analyses, four independent solvers. Everything numeric on this sheet is Gen5.
Current direction	stage-integrated, gas-driven	the payload accelerated directly along a rail a spent upper stage already provides. It has not inherited Gen5's validation and nothing here is claimed for it.

THE CONCEPT OF OPERATIONS



The host is an active participant, not a mounting surface. A 100 m/s host budget reaches four shells and a fleet spanning 269 km of altitude; plane change is excluded at 133 m/s per degree. The altitude half of any multi-orbit claim is bought with host propellant, not with this machine.

GEN5, AT THE RATED POINT

Thrust constant	10.54 N per kA/m	Exit velocity, 3U	16.029 m/s
ripple	±1.01 %	acceleration	10.07 g for 162 ms
2-D FEM agrees	to 0.03 %	Dispersion, closed loop	0.0274 m/s (3σ)
3-D FEM agrees	to 0.059 %	as apogee	±0.10 km
Energy per shot	2782 J gross	Semi-major axis change	+28.8 km
net of recovery	2735 J	Orbital lifetime	× 1.60 at mean activity
to the payload	514 J — 18.8 %	against a spring	+60.2 % vs +8.2 %
Pulse	162.3 ms, 320 A peak	Mass, dry / loaded	126.6 / 174.6 kg
Track first mode	109.0 Hz	per 3U satellite	10.55 kg
Recoil per shot	64.1 N·s	Recurring cost	₹ 1,345,055 — every price assumed

Every figure above is a model output, single-sourced to a committed JSON result and re-checked against its script on every commit. **None of it is a measurement, and no vendor has quoted any line of the cost.**

The contribution is the combination, not any one part of it. Electromagnetic launch of CubeSats has been studied before, at velocities twenty times higher — but at accelerations of order $10^3 g$ and with a conductive armature bonded to the payload. What is claimed here is a **commanded per-satellite velocity, on a satellite that is never modified, at two orders of magnitude less acceleration, from a stage that was going to be discarded.** Together they describe a service tier with no demonstrated occupant. **Whether a given satellite tolerates 10.07 g is not claimed:** 25 g is a ceiling this design sets on itself, no standard fixes a universal quasi-static limit for CubeSats, and compatibility is a payload-specific structural question needing an integration review that has not been done.

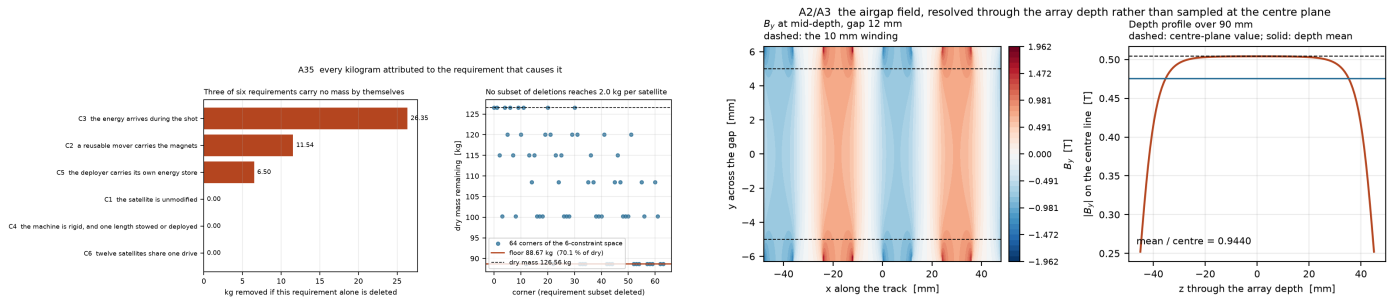
How the numbers were made, and what they cost the argument

THE METHOD CHAIN

Geometry	Parametric CAD built from one parameter file; 23 dimensions read back out of the model and checked against it.
Electromagnetics	Analytic travelling-wave model, cross-checked by magpylib , then by a 2-D magnetostatic FEM written from the weak form (scikit-fem, 141 k elements) to 0.03 %, then by a 3-D reduced-scalar-potential solve (getdp, 274 105 DoF) to 0.059 %.
Structures	CalculiX on the sled chassis and the retention chain.
Aerodynamics	OpenFOAM simpleFoam external solve, 581 779 cells – for the air correction a ground test would need, not for flight.
Circuit	ngspice transient on the pulse chain, against the analytic ledger.
Astrodynamics	GMAT R2022a headless, MSISE90, 20×20 gravity, lunisolar and SRP, against this project’s own propagator.
Control	Closed-loop Monte Carlo, 800 runs, with modelled sensing.
The discipline	64 run-sheet files covering 61 analyses. Each declares its acceptance bands, and commits them, before the script that produces the number exists – provable by checking that the band commit touched no script. A failed band moves the design point. No band has ever been widened after a result was known.

WHAT THE ANALYSIS CHANGED – THE FOUR THAT COST THE MOST

- 1 **Exact CAD mass took the headline from 20.37 to 16.029 m/s.** A parametric 4.86 kg sled became a 9.445 kg one when solid volumes were computed; resolving the field through the array’s 90 mm depth, rather than sampling the centre plane, then took another 4.42 %. *Both consequences were declared in writing before the analyses ran.*
- 2 **An independent propagator falsified a claim in this project’s own abstract.** The orbital-lifetime ratio had been offered as invariant across ballistic coefficient and solar activity. GMAT disagreed at low activity by 18 % against a 5 % band. The cause was the sweep, not the arithmetic: both variables entered the same multiplicative slot, so the ratio could not have moved. *The claim is withdrawn in the manuscript.*
- 3 **ngspice found a term no script carried.** Energy closure failed at 97.0 % against a declared 98–102 %. The missing **86.6 J a shot** was the capacitor bank’s own series resistance, which the analytic ledger had no term for at all.
- 4 **Every kilogram was attributed to the requirement causing it, and then every requirement was deleted.** Across all **64 corners**, **88.67 kg survives** – 70.06 % of dry mass, **7.39 kg per satellite against a 2 kg criterion.** *This is why Gen5 cannot be optimised into its own mass target, and it is the single result that redirected the work.*



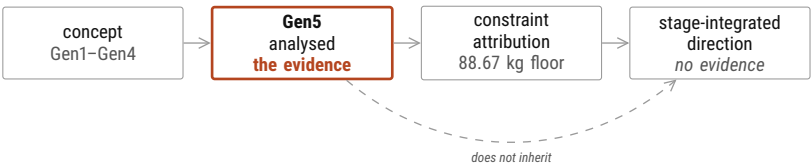
Left: the constraint ledger – what each requirement costs alone, and the floor no combination of deletions reaches. Right: the airgap field, and the depth profile that made the thrust constant 4.42 % smaller. Both are model outputs.

THE HONEST COMPARISON

	Spring dispenser	VOLLEY Gen5	Cold-gas module on the satellite
Δv per satellite	1–2.5 m/s, one value for all	16.03 m/s, commanded	~16 m/s
Designed differential	exactly 0	16.03 m/s, to 0.027 (3σ)	per satellite
Orbital energy	+8.2 % life at 2.5 m/s	+28.8 km, ×1.60 life	comparable
Mass per 3U sat.	~6.0 kg	10.55 kg – loses, 1.758 ×	0.5–1.2 kg – loses, 12.4 ×
Satellite carries	nothing	nothing	pressure vessel, qualification, range safety
Host provides	one deploy signal	150–300 W, serial link, firing window	nothing
Maturity	TRL 9, thousands flown	TRL 2–3, nothing measured – loses	COTS, flown

Two of the three rows that matter commercially are losses, and they are printed because they are the result. Mass parity with a canister of springs was claimed by this project and is **withdrawn**: an acceptance band asked for parity within 15 % and the measured ratio is 1.758. So is the constellation-phasing claim – 30° of in-track phase costs **468 seconds of waiting** at zero Δv , so a spring and a clock reach it and this machine is not needed for it. **What survives is the third row: a clock changes phase; a commanded deployment impulse changes orbital energy.** Drag, J_2 and solar pressure change orbits too – a clock is not the only thing that cannot. What a clock cannot produce is the **+28.8 km of semi-major axis** a commanded shot does, and no spring at any preload produces it either. No cost comparison is offered in either direction; there is no quotation for any of it.

WHERE IT GOES NEXT



The next step is not more analysis. It is a benchtop track and a mass simulator – the point at which the *measured* evidence class in this project’s own figure index stops having zero members.



The full record
github.com/
aaaaaaaaaaaavm/VOLLEY

Status. Phase I design study, TRL 2–3. Nothing has been built, fired, measured, qualified or flown, and no result here has been reviewed by a third party. Model-to-model agreement between two solvers is a consistency check, not experimental validation. The defect register carries **133 numbered entries, 53 of them still open**, including every one that damages the claims above. **No institutional, agency or commercial affiliation is claimed or implied by this document.**